

Bahman Gharesifard
Curriculum vitae: September, 2026

Professor

Department of Mathematics and Statistics
Queen's University

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Please visit this link for a complete version of my CV: <https://gharesifard.github.io/pdfs/CV.pdf>

POSITIONS

Professor of Mathematics (with tenure), Queen's University July 2024 –
Department of Mathematics and Statistics

Professor (with tenure), University of California, Los Angeles July 2021– July 2024
Area Director for Signals and Systems July 2023– July 2024
UCLA Samueli School of Engineering
Department of Electrical and Computer Engineering

Associate Professor, Queen's University July 2019 – June 2021
Department of Mathematics and Statistics

Alexander von Humboldt Fellow, University of Stuttgart October 2019 – December 2020
Institute for Systems Theory and Automatic Control

Ontario/Baden-Württemberg Visiting Faculty, University of Stuttgart Summer 2016
Institute for Systems Theory and Automatic Control

Assistant Professor, Queen's University July 2013 – July 2019
Department of Mathematics and Statistics

Postdoctoral Research Associate, University of Illinois, Urbana-Champaign March 2012 – July 2013
Coordinated Science Laboratory
Department of Electrical and Computer Engineering

Postdoctoral Research Associate, University of California San Diego September 2009 – March 2012
Jacobs School of Engineering
Department of Mechanical and Aerospace Engineering

EDUCATION

Ph.D., Mathematics
Queen's University, Kingston, ON, Canada October 2009

B.Sc. and M.Sc., Mechanical Engineering (Dynamics and Control)
Shiraz University, Shiraz, Iran May 2005

RESEARCH INTERESTS

Decision Theory: systems and control; stability of discrete-time/continuous-time dynamical systems; networked control systems; learning for control; stochastic control; sensor networks; discontinuous dynamical systems; nonlinear and geometric control; autonomy; controllability of partial differential equations; economic dispatch of energy in networked power systems; identification and adaptation

Machine Learning and Optimization Theory: mathematical theory of neural networks, and transformers; sample complexity analysis; reinforcement learning; decentralized optimization; statistical learning; submodular optimization; online learning; GANs; model collapse in generative models

Optimal Transport: intersections with control, learning, game theory, and optimization; continuity equations

Social Networks: opinion dynamics; diffusion models; reinforcement schemes; random graphs

Game Theory and Economics: game theory and strategic interactions; hypergames; market design in distributed settings; learning in games

Stochastic Processes and Probability Theory: stochastic reinforcement processes, stochastic differential equations and control; rough path theory

Global Analysis: applied differential geometry; exterior differential systems; Lie theory, Riemannian geometry; geometric theory of partial differential equations

AWARDS AND HONORS

Invited Speaker , Mathematical Foundations of AI and Data-Driven Scientific Computing, National University of Singapore, Singapore	2026
Semi-plenary Speaker , 27th International Symposium on Mathematical Theory of Networks and Systems University of Waterloo	2026
The 2024 CSS TC on Robust and Complex Systems Outstanding Student Paper Prize	2024
Canadian Society of Information Theory Best Paper Award	2022
Best conference paper that has been accepted by either the Biennial Symposium on Communications (BSC) or the Canadian Workshop on Information Theory (CWIT)	
SIAG/CST Best SICON Paper Prize 2021	2021
Awarded to the author(s) of the two most outstanding papers, as determined by the prize committee, published in the SIAM Journal on Control and Optimization (SICON) in the three calendar years before the award year	
Booz Allen Hamilton Distinguished Colloquium Lecture , University of Maryland	2020
Alexander von Humboldt Fellowship for experienced researchers	2019
The 2019 CAIMS-PIMS Early Career Award	2019
Jointly awarded by the Canadian Applied & Industrial Math Society and the Pacific Institute for the Mathematical Sciences	
NSERC Discovery Accelerator Supplement	2019
supplemental funding with 125 recipients among over 3000 applicants in Canada	
Finalist for Best Paper Award (as a supervisor), American Control Conference	2017
Class of 2020 Engineering and Applied Science First Year Instructor Teaching Award	2017
Class of 2018 Engineering and Applied Science First Year Instructor Teaching Award	2015
Shortlisted for the Frank Knox Award for Excellence in Teaching	2014

SELECTED RECENT PUBLICATIONS

Publications sorted by research areas: <https://gharesifard.github.io/publications.html>

1. Alvaro Rodriguez Abella, Joao Silvestre, *Bahman Gharesifard*, and Paulo Tabuada, Deep learning from a control vantage point, **Control Systems Magazine**, to appear, 2026 ([Download](#))
2. *Bahman Gharesifard*, On the topology of neural network superlevel sets, **Comptes Rendus, Série Mathématique**, submitted, 2026 ([arXiv](#))
3. Paulo Tabuada and *Bahman Gharesifard*, Universal approximation power of deep residual neural networks through the lens of control, **IEEE Transactions on Automatic Control**, and **ICLR**, 2023 ([arXiv](#))
4. Matteo Marchi, Johnathan Bunton, *Bahman Gharesifard*, and Paulo Tabuada, Stability guarantees for control loops with deep learning state estimation, **IEEE Control Systems Letters**, 2021 ([Link](#))
5. *Bahman Gharesifard*, Matteo Marchi, Jonathan Bunton, Paulo Tabuada, Preventing model collapse: A Fisher-Rao perspective on the dynamics of training with synthetic data, **Proceedings of the IEEE Conference on Decision and Control**, Honolulu, to appear, 2026 ([Download](#))
6. *Bahman Gharesifard* and Paulo Tabuada, Preventing model collapse when training LLMs with synthetic data, **Proceedings of the IEEE Conference on Decision and Control**, Rio de Janeiro, Brazil, 2025 ([Link](#))
7. Shamak Dutta, *Bahman Gharesifard*, and Stephen Smith, Submodular optimization with applications to decision and control, **Automatica**, submitted, 2026 ([arXiv](#))
8. Michela Lapenna, Rita Fioresi, and *Bahman Gharesifard*, Sinkhorn doubly stochastic attention rank decay analysis, **Transactions on Machine Learning Research**, conditionally accepted, 2026 ([arXiv](#))
9. Arijit Dey and *Bahman Gharesifard*, Graph State Networks (GSNs): persistent node-wise selective State Space Models, **Transactions on Machine Learning Research**, 2026 ([Link](#))
10. Simone Baroncini, *Bahman Gharesifard*, and Giuseppe Notarstefano, On reward-balancing in Reinforcement Learning, **IEEE Transaction on Automatic Control**, submitted, 2026 ([arXiv](#))
11. Amirreza Neshaei, Alex Olshevsky, and *Bahman Gharesifard*, Sample complexity of the linear quadratic regulator: a reinforcement learning lens, **Journal of Machine Learning Research**, 2025 ([Link](#))
12. Henri Cimetiere, Maria Teresa Chiri, and *Bahman Gharesifard*, Localmax dynamics for attention in transformers and its asymptotic behavior, **SIAM Journal on Control and Optimization**, submitted, 2025 ([arXiv](#))
13. Daniel Adu and *Bahman Gharesifard*, Approximate controllability of continuity equation of transformers, **IEEE Control Systems Letters**, 2024 ([Link](#))
14. Alex Olshevsky and *Bahman Gharesifard*, A small-gain analysis of single timescale actor critic, **SIAM Journal on Control and Optimization**, 2023 ([Link](#))
15. Karthik Elamvazhuti, *Bahman Gharesifard*, Andrea Bertozzi, and Stanley Osher, Neural ODE control for trajectory approximation of continuity equation, **IEEE Control Systems Letters**, 2022 ([Link](#))